Sensorless Control of Low Speed PMSM Based on Novel Sliding Mode Observer

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***Abstract*—This paper presents a novel sliding mode observer (SMO) based on the permanent magnet flux linkage of permanent magnet synchronous motor (PMSM). The stability of the proposed SMO is further proved with the Lyapunov function. As the sliding mode gain required by the proposed SMO to satisfy the stability condition is much smaller than the traditional back-electromotive force based SMO, the proposed SMO effectively weakens chattering and avoids the use of low-pass filter, delivers higher precision of position and speed estimation in low speed. A linear extended state observer (LESO) is incorporated into the quadrature phase locked loop (Q-PLL) to achieve better dynamic performance under large load torque. Finally, the proposed sensorless control strategy is tested in MATLAB/Simulink for its speed tracking capability under low speed. Simulation results validate the feasibility and effectiveness of the proposed sensorless control strategy.**

Keywords*—*sliding mode observer (SMO); permanent magnet flux linkage; permanrnt magnet synchronous motor (PMSM); linear extended state observer; low speed;

I. Introduction

Permanent magnet synchronous motors (PMSMs) are widely used in industrial applications due to its high power density, high efficiency, high torque-current ratio [1]. PMSMs need the rotor position signal to realize the synchronization of the rotating magnetic field of the stator and the rotor, the rotor position signal needs to be obtained by a high-precision sensor installed on the rotor shaft [2-4]. Such sensor presents some disadvantage such as increase drive cost, size and reduce reliability, thus limiting the use of PMSMs in specific situations. Therefore, sensorless control without position and speed sensors are increasingly becoming a popular research topic in literature [5-6].

The existing sensorless control methods can be divided into two categories. The first category is the high-frequency signal injection method based on PMSM anisotropy, and the second type is the back electromotive force (back-EMF) method based on the fundamental frequency mathematical model of the PMSM [7]. Among all kinds of sensorless control solutions, sliding mode observers (SMO) based on back-EMF are widely used because of its robustness and ease of implementation [8]. Existing papers have conducted much research on the sensorless control strategy of the sliding mode observer based on the back-EMF and the quadrature phase locked loop (Q-PLL). However, since the amplitude of the back-EMF is proportional to the electrical angular velocity of the rotor. Therefore, when the motor runs at medium and high speeds, the back-EMF is relatively large, thus the accuracy of the extracted rotor position information is high, while when the motor runs at low speed and the start-up stage, the back-EMF is small, the signal-noise ratio is low, and valid rotor position information cannot be observed [9]. And the Q-PLL, used for extracting speed and position information, contains a linear PI regulator as error correction mechanism. Thus causes some problems, such as a contradiction between rapidity and stability and less capability of dealing with time-varying perturbation.

In this paper, a SMO based on the permanent magnet flux linkage for PMSM is proposed, compared with the traditional SMO based on back-EMF, the novel SMO proposed in this paper can provide more accurate rotor position information at low speed and start-up stage, and a LESO based Q-PLL is proposed to achieve fast response and good dynamic performance under large disturbances. Finally, the effectiveness of the proposed sensorless control strategy is verified by simulation results.

II. Proposed Sliding Mode Observer

A. Mathematic Model of PMSM

The mathematical model of the surface mounted permanent magnet synchronous motor (SPMSM) in the stationary frame can be written as

$$\begin{bmatrix} \dfrac{di_\alpha}{dt} \\ \dfrac{di_\beta}{dt} \end{bmatrix} = \begin{bmatrix} \dfrac{1}{L_s} & 0 \\ 0 & \dfrac{1}{L_s} \end{bmatrix} \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - \begin{bmatrix} \dfrac{R_s}{L_s} & 0 \\ 0 & \dfrac{R_s}{L_s} \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} - \begin{bmatrix} \dfrac{1}{L_s} & 0 \\ 0 & \dfrac{1}{L_s} \end{bmatrix} \begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} \quad (1)$$

where u_α, u_β are stator voltages component in $\alpha\beta$-axes; i_α, i_β are stator currents component in $\alpha\beta$-axes; R_s is the stator resistance; L_s is the stator inductance; e_α and e_β are the $\alpha\beta$-axes back-EMF defined as

$$\begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} = \omega_e \psi_f \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \quad (2)$$

where ψ_f is the permanent magnet flux linkage; ω_e is the rotor electrical rotor speed; θ_e is the electrical position of rotor.

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B. Design of Novel Sliding Mode Observer

As the stator current is easy to measure, the sliding mode surface is defined as

$$S(x)=\begin{bmatrix} S_\alpha \\ S_\beta \end{bmatrix}=\begin{bmatrix} \frac{R_s}{L_s}\int \tilde{i}_\alpha dt+\tilde{i}_\alpha \\ \frac{R_s}{L_s}\int \tilde{i}_\beta dt+\tilde{i}_\beta \end{bmatrix} \tag{3}$$

where $\tilde{i}_\alpha=\hat{i}_\alpha-i_\alpha$, $\tilde{i}_\beta=\hat{i}_\beta-i_\beta$, $\hat{i}_\alpha$ and $\hat{i}_\beta$ represents the estimated value of i_α and i_β ,respectively. The integral in the sliding mode surface can eliminate steady-state errors and effectively suppress chattering problems.

Construct the sliding mode observer as follows

$$\begin{bmatrix} \frac{d\hat{i}_\alpha}{dt} \\ \frac{d\hat{i}_\beta}{dt} \end{bmatrix}=\begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \end{bmatrix}\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix}+\begin{bmatrix} -\frac{R_s}{L_s} & 0 \\ 0 & -\frac{R_s}{L_s} \end{bmatrix}\begin{bmatrix} \hat{i}_\alpha \\ \hat{i}_\beta \end{bmatrix}-\begin{bmatrix} \frac{\omega_e}{L_s} & 0 \\ 0 & \frac{\omega_e}{L_s} \end{bmatrix}\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix} \tag{4}$$

$$\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix}=k\begin{bmatrix} sign(S_\alpha) \\ sign(S_\beta) \end{bmatrix} \tag{5}$$

where sign represents the switch function; k is the sliding mode gain of switching function.

Subtracting (4) into (1)

$$\begin{bmatrix} \frac{d\tilde{i}_\alpha}{dt} \\ \frac{d\tilde{i}_\beta}{dt} \end{bmatrix}=\begin{bmatrix} -\frac{R_s}{L_s} & 0 \\ 0 & -\frac{R_s}{L_s} \end{bmatrix}\begin{bmatrix} \tilde{i}_\alpha \\ \tilde{i}_\beta \end{bmatrix}+\begin{bmatrix} \frac{1}{Ls} & 0 \\ 0 & \frac{1}{Ls} \end{bmatrix}\begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix}-\begin{bmatrix} \frac{\omega_e}{L_s} & 0 \\ 0 & \frac{\omega_e}{L_s} \end{bmatrix}\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix} \tag{6}$$

When reaching the sliding mode surface, $\tilde{i}_\alpha=0$,thus (6) can be simplified as

$$\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix}=k\begin{bmatrix} sign(S_\alpha) \\ sign(S_\beta) \end{bmatrix}=\begin{bmatrix} -\psi_f \sin\theta_e \\ \psi_f \cos\theta_e \end{bmatrix} \tag{7}$$

From (7), it is clear that z_α, z_β contains the information of rotor position. Therefore, the information of the rotor position can be obtained by Q-PLL.

It is worth mentioning that the back-EMF based SMO is unable to extract accurate rotor position information when motor runs at low speed and start-up stage, since the back-EMF is small, the signal-noise ratio is low. While the permanent magnet flux linkage is a fix value, thus it does not change with speed. Therefore, the SMO proposed in this paper can still accurately extract rotor position information at low speed and start-up stage.

C. Stability Analysis

The Lyapunov function used to find the sliding condition is defined as

$$V=\frac{1}{2}S^T S \tag{8}$$

The observer is stable only if $\dot{V}<0$ for $V>0$.

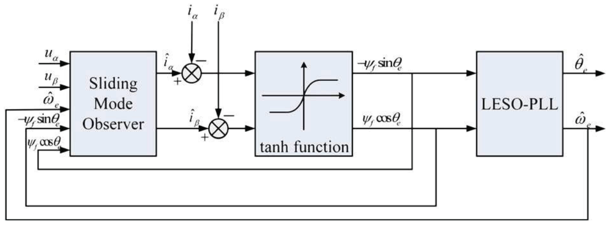

Fig. 1. Block diagram of the proposed novel sliding mode observer.

$$\begin{aligned}\dot{V}&=S^T\dot{S}\\ &=(\frac{R_s}{L_s}\int\tilde{i}_\alpha \mathrm{dt}+\tilde{i}_\alpha)(\frac{R_s}{L_s}\tilde{i}_\alpha+\dot{\tilde{i}}_\alpha)+(\frac{R_s}{L_s}\int\tilde{i}_\beta \mathrm{dt}+\tilde{i}_\beta)(\frac{R_s}{L_s}\tilde{i}_\beta+\dot{\tilde{i}}_\beta)\\ &=\mathrm{S}_\alpha(\frac{R_s}{L_s}\tilde{i}_\alpha+\dot{\tilde{i}}_\alpha)+\mathrm{S}_\beta(\frac{R_s}{L_s}\tilde{i}_\beta+\dot{\tilde{i}}_\beta)\end{aligned} \tag{9}$$

Combine (9) with (6), the following equation can be established:

$$\begin{aligned}\dot{V}&=S_\alpha(\frac{1}{L_s}e_\alpha-\frac{w_e}{L_s}\mathrm{k}\cdot\mathrm{sign}(S_\alpha))+S_\beta(\frac{1}{L_s}e_\beta-\frac{w_e}{L_s}\mathrm{k}\cdot\mathrm{sign}(S_\beta))\\ &=\frac{e_\alpha\cdot S_\alpha-w_e\cdot k|S_\alpha|+e_\beta\cdot S_\beta-w_e\cdot k|S_\beta|}{L_s}\\ &\le\frac{|w_e|\psi_f\cdot S_\alpha-w_e\cdot k|S_\alpha|+|w_e|\psi_f\cdot S_\beta-w_e\cdot k|S_\beta|}{L_s}\end{aligned} \tag{10}$$

Therefore, the sliding mode condition to satisfy $\dot{V}<0$ is $k>\psi_f$. It is worth mentioning that in traditional SMO the sliding mode condition to satisfy Lyapunov function is $k>\max(|e_\alpha|,|e_\beta|)$. In most cases, $\psi_f \ll \max(|e_\alpha|,|e_\beta|)$.Thus the SMO proposed in this paper weakens chattering problems and reduces the dependence on low-pass filter. To further suppress chattering problem, the sign function can be replaced by a tanh function or a sigmoid function. The proposed block diagram of the novel SMO is illustrated in Fig. 1.

III. Linear Extended State Observer Based Q-PLL

A. Traditional Q-PLL

Q-PLL has the advantages of simple structure and easy implementation, thus Q-PLL is widely used to extract the position information in the back-EMF obtained in the traditional SMO. The structure diagram of Q-PLL is shown in Fig. 2.

The closed-loop transfer function of the traditional Q-PLL is shown in (11)

$$G_{PLL}(s)=\frac{\omega_e\psi_f(K_p s+K_i)}{s^2+\omega_e\psi_f K_p s+\omega_e\psi_f K_i} \tag{11}$$

From (11), it is clear that the cut-off frequency of the traditional PLL is affected by the electrical angular velocity of the motor. This problem can be solved by orthogonalization, but orthogonalization increases the amount of calculation. And the Q-PLL contains a linear PI regulator as error correction mechanism. Thus causes some problems, such as a contradiction between rapidity and stability and less capability of dealing with time-varying disturbance.

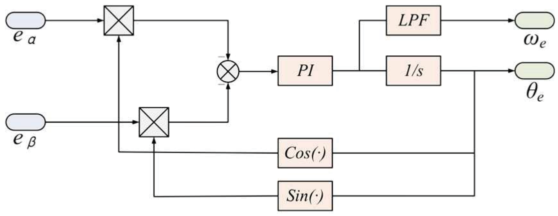

Fig. 2. Block diagram of the traditional Q-PLL.

B. Linear Extended State Observer Based Q-PLL

The state equation of rotor angular speed and rotor position are represented by

$$\begin{cases} \dot{\theta}_e = \omega_e \\ \dot{\omega}_e = \dfrac{1.5P_n^{\,2}\psi_f i_q - P_n T_l - B\omega_e}{J} \end{cases} \tag{12}$$

where ω_e is the rotor electrical angular velocity; θ_e is the electrical position of rotor; J is the moment of inertia; P_n is pole pairs; T_l is the load torque; B is viscous friction coefficient.

The following extended state observer is established for the above state equation

$$\begin{cases} e = \hat{\theta}_e - \theta_e \\ \dot{\hat{\theta}}_e = \hat{\omega}_e - \beta_{01} e \\ \dot{\hat{\omega}}_e = \dfrac{1.5P_n^{\,2}\psi_f i_q - B\omega_e}{J} + f - \beta_{02} e \\ \dot{f} = -\beta_{03} e \end{cases} \tag{13}$$

$$\begin{cases} \beta_{01} = 3\omega_0 \\ \beta_{02} = 3\omega_0^{\,2} \\ \beta_{03} = \omega_0^{\,3} \end{cases} \tag{14}$$

where ω_0 is the observer bandwidth, the observer bandwidth can be deduced using the pole placement approach [10]. Small observer bandwidth will result in poor tracking performance, while large observer bandwidth will result in the increase of noise sensitivity. The observer bandwidth is generally set to three to five times the controller bandwidth.

The block diagram of the LSEO based Q-PLL proposed in this paper is shown in Fig. 3.

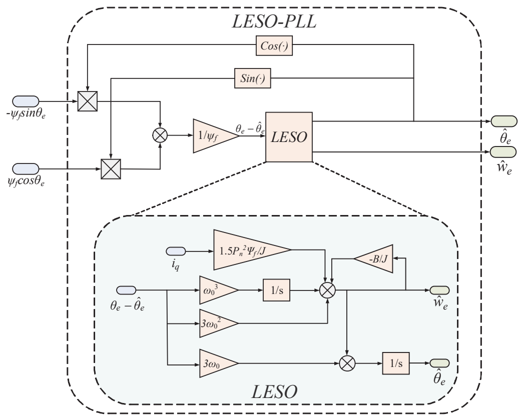

Fig. 3. Block diagram of the LESO based Q-PLL.

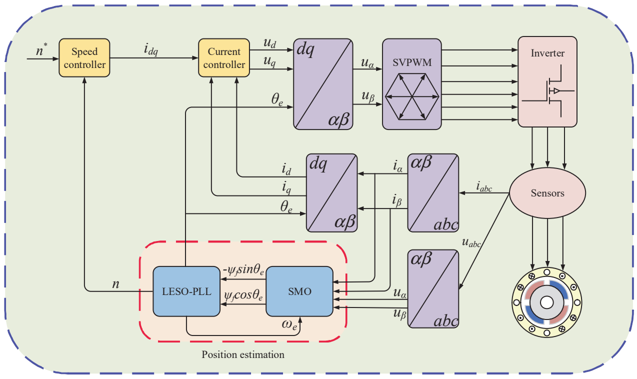

Fig. 4. Block diagram of the proposed sensorless PMSM drive system.

It is worth mentioning that the LSEO based Q-PLL proposed in this paper avoids the use of low-pass filter, and the cut-off frequency of the system will not be affected by the electrical angular velocity of the motor.

IV. Simulation Result And Analysis

The simulation diagram of the proposed sensorless PMSM drive system is shown in Fig. 4, simulation is performed based on MATLAB/Simulink. The simulation step size is set as a fixed step size of 0.00001s. All the PMSM parameters are given in Table I. As shown in (10), the sliding mode gain is set as 6.

TABLE I. Parameters Of Pmsm

Parameters	***Value***
Rated power(kW)	210
Rated voltage(V)	1140
Rated speed(r/min)	90
Pole pairs	20
Stator resistance(Ω)	0.11873
Stator inductance(mH)	5.555
Flux(Wb)	5.19
Moment of inertia(kg·m²)	236.8

In the simulation, the reference speed is 30r/min, 60r/min and 90r/min, the comparison between the real speed and the estimated speed is shown in Fig. 5, and the speed estimation error is shown in Fig. 6.

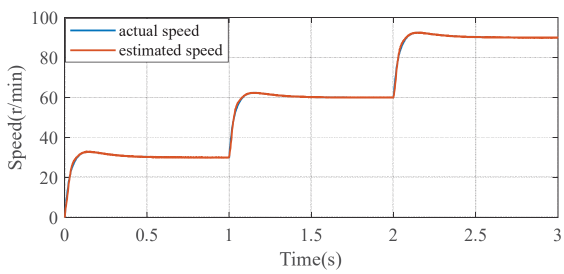

Fig. 5. Simulation of the actual and estimated speed.

It can be seen from Fig. 5 that the novel SMO proposed in this paper has fast dynamic response and can accurately track the actual speed at low speed. When the motor is running in steady state, the estimated error of speed remains within 0.1r/min. When the given speed changes, the error will increase, but it is completely within the acceptable range. And when the motor runs in steady state again, the estimated error of speed remains within 0.1r/min again, and it can meet industry requirements.

Fig. 7 shows the estimated and actual value of rotor position. Fig. 8 shows the estimated error of rotor position. It can be seen from Fig. 7 and Fig. 8 that when the motor is running in steady state, the angle error remains within 0.003 rad.

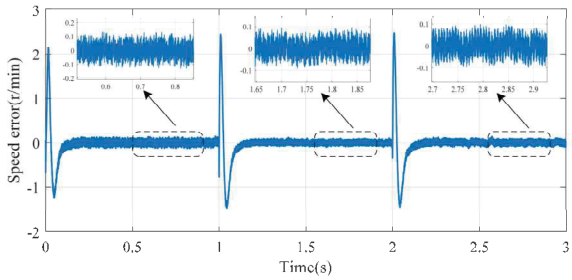

Fig. 6. Simulation of the estimated error of speed.

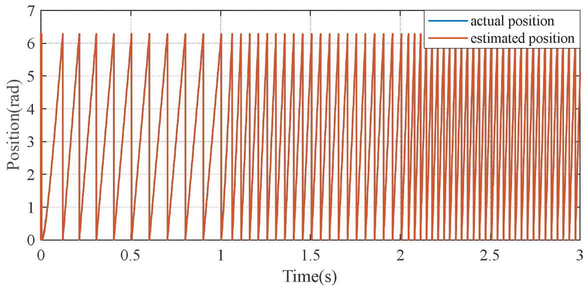

Fig. 7. Simulation of the estimated and actual value of rotor position.

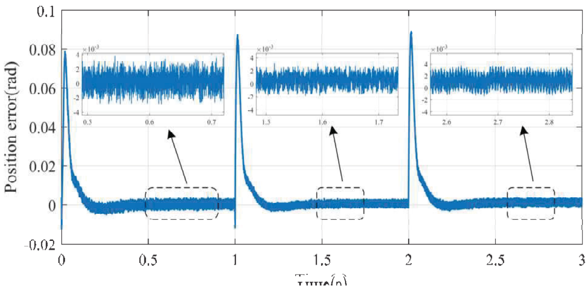

Fig. 8. Simulation of the estimated error of rotor position.

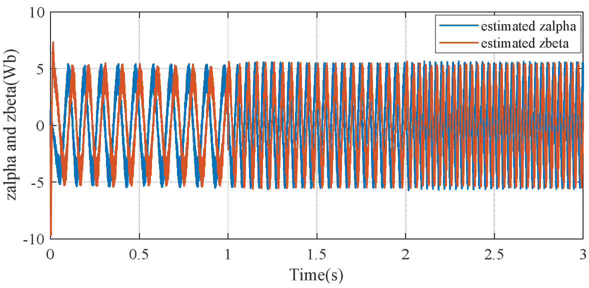

Fig. 9. Simulation of the estimated zalpha and zbeta.

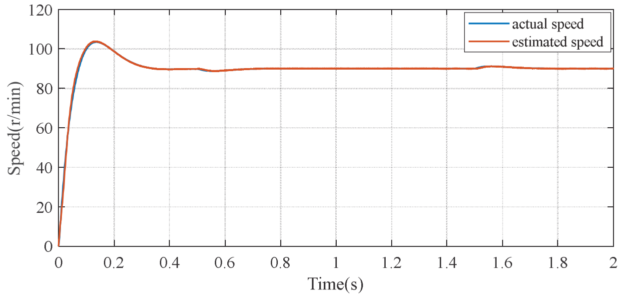

Fig. 10. The estimated and actual value of rotor speed with large load torque.

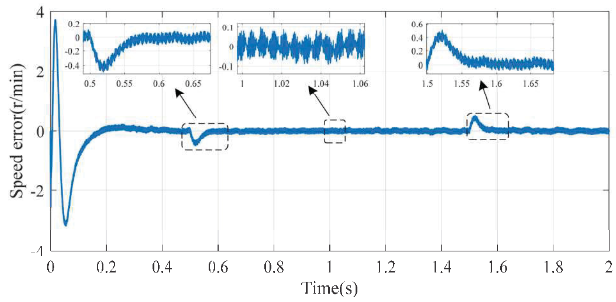

Fig. 11. The estimated error of speed with large load torque.

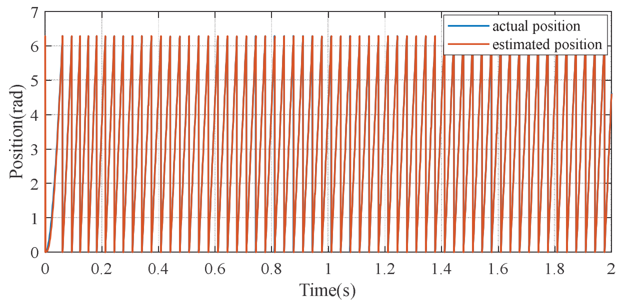

Fig. 12. The estimated and actual value of rotor position with large load torque.

Fig. 9 shows the estimated values of Z_α and Z_β. The equation of Z_α and Z_β is shown in (7). It can be seen from Fig. 9 that under the combined effect of a small sliding mode gain and the integral sliding mode surface, the Z_α and Z_β can still maintain a good sinusoidal shape without using a low-pass filter.

Fig. 10 to Fig. 13 shows the simulation results of the proposed sensorless control strategy under large load torque condition. The PMSM starts with a load torque of 3000N·m, the load torque changes to 4000N·m at 0.5s, and the load torque change to 3000N·m at 1.5s.

The comparison between the actual speed and the estimated speed with large load torque is shown in Fig. 10, and the estimated error of speed with large load torque is shown in Fig. 11. It can be seen from Fig. 11 that estimated error of speed during the start-up stage is within 4r/min. And when the reference speed changes suddenly, the estimated error of speed will increase, but the estimated error is within 0.5r/min, which is completely acceptable in engineering. And when the motor is running in steady state again, the estimated error of speed remains within 0.1r/min.

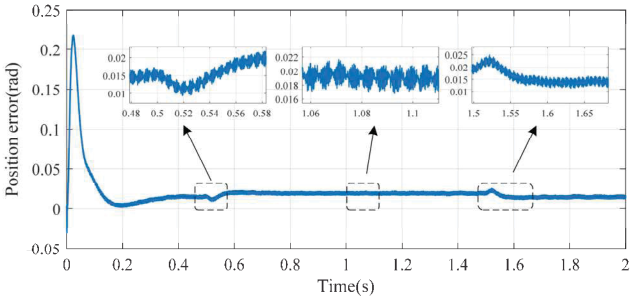

Fig. 13. The estimated error of rotor position with large load torque.

The comparison between the actual position and the estimated position with large load torque is shown in Fig. 12, and the estimated error of rotor position with load is shown in Fig. 13. It can be seen from Fig. 12 and Fig. 13 that under large load torque condition, the sensorless control strategy proposed in this paper can still track the rotor position accurately, and the estimated position error remains within 0.021 rad.

V. CONCLUSION

This paper proposes a novel sliding mode observer based on the permanent magnet flux linkage of PMSM. The proposed novel SMO can provide accurate rotor position at low speed and start-up stage. A linear extended state observer is incorporated into the traditional quadrature phase-locked-loop to further improve the dynamic response performance under large load torque condition. The proposed sensorless control strategy is tested based on MATLAB/Simulink. Simulation results have shown that the proposed sensorless control strategy can accurately track the rotor position and speed under low speed and start-up stage. Further simulations are tested under large load torque and time-varying load torque conditions, the simulation results show that the proposed sensorless control strategy can also provide accurate rotor position information under large load torque and time-varying load torque conditions.

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